

What is the first cell that this code colors?

```
int rows = g.getNumRows();
int cols = g.getNumCols();
for(int i=0; i<rows; i++) {
    for(int j=0; j<cols; j++) {
        g.setColor(i, j, Color.RED);
    }
}
```

A																B
C																D

E: None of the above

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```

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What is the second cell that this code colors?

```
int rows = g.getNumRows();
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for(int i=0; i<rows; i++) {
    for(int j=0; j<cols; j++) {
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    }
}
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	A															
B	C															
																D

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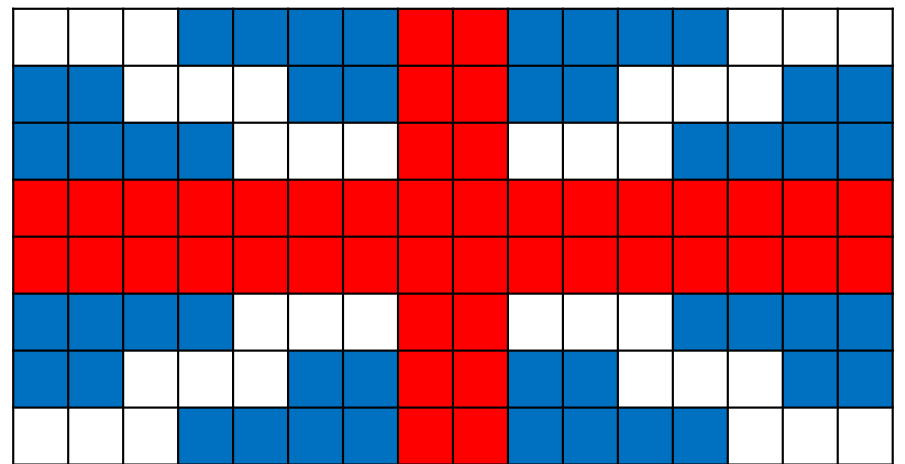
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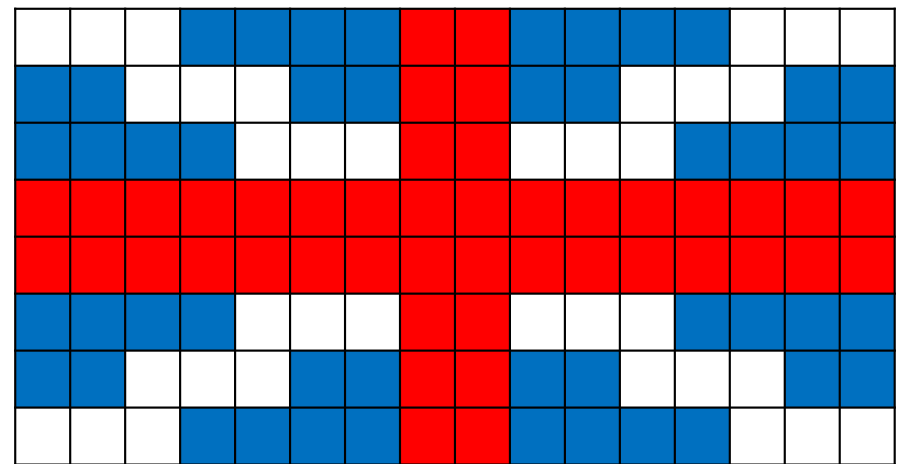
How would you draw the British flag?



[https://en.wikipedia.org/wiki/Flag_of_Great_Britain#/media/File:Flag_of_Great_Britain_\(1707%E2%80%931800\).svg](https://en.wikipedia.org/wiki/Flag_of_Great_Britain#/media/File:Flag_of_Great_Britain_(1707%E2%80%931800).svg)

One strategy: Drawing “layers”

- Color entire background blue
- Draw white diagonals
- Draw vertical and horizontal red lines



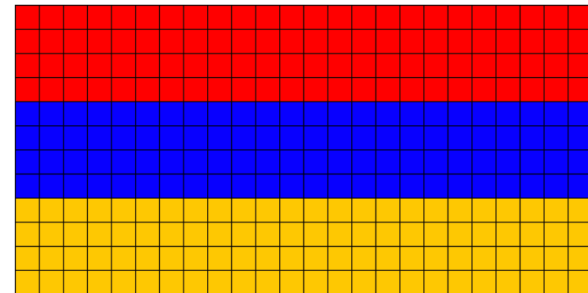
Advanced topic: Parallel computation

(concept only now, do in later courses)

- Modern computers are all multicore (dual-core, quad-core, ...) so they can perform multiple operations at a time
- Potentially speeds up some tasks, like drawing the Armenian flag

Simultaneously do these tasks:

- Draw red stripe
- Draw blue stripe
- Draw yellow stripe



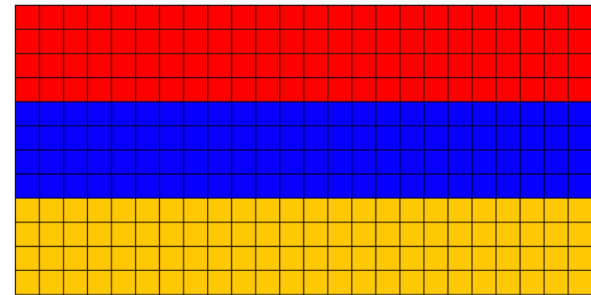
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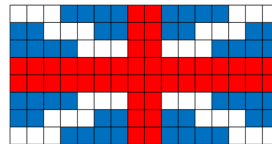
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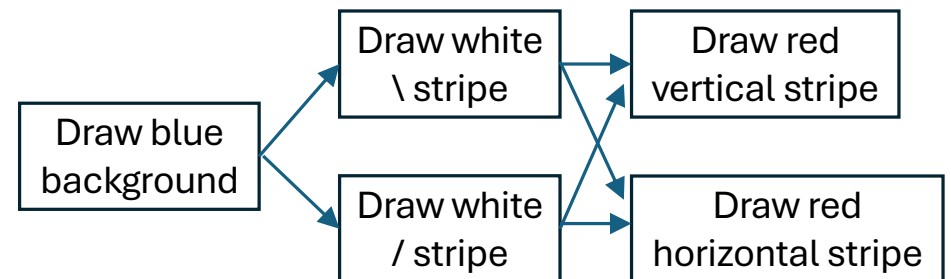
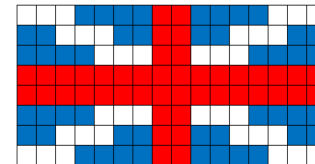
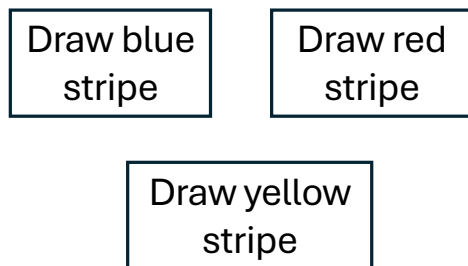
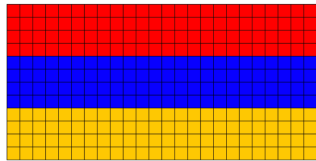


Can we do this with the British flag?

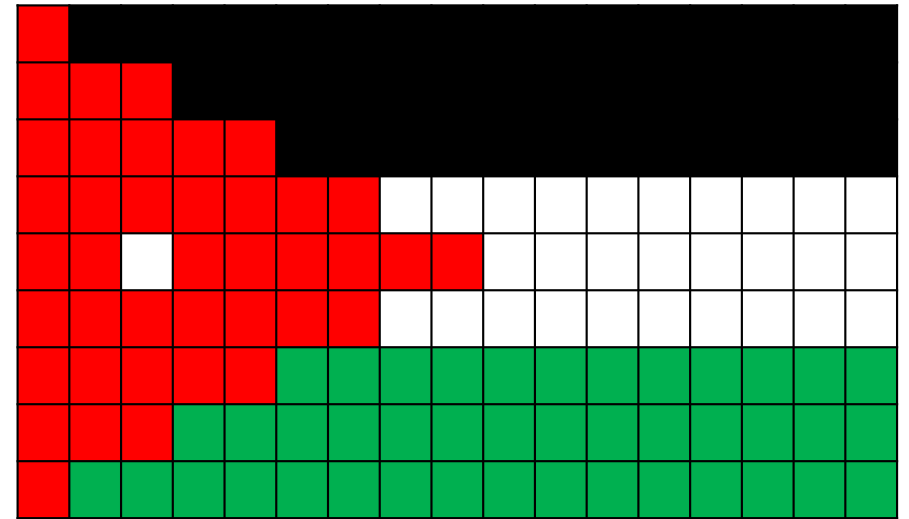


“Task Graph”

- Diagram of dependencies (when one thing must come before another)
 - For flags: when operations would color a tile different colors
 - Draw arrows to show which must come before others



Create a task graph to draw the flag of Jordan



https://upload.wikimedia.org/wikipedia/commons/c/c0/Flag_of_Jordan.svg